Imported Rates and Domestic Amplification: BVAR Evidence from Hong Kong's Currency Board

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Abstract

Hong Kong's Linked Exchange Rate System (LERS) eliminates the exchange rate channel of monetary transmission, providing an institutional setting in which imported interest rate conditions reach the domestic economy through financial and trade channels. This paper estimates a Bayesian VAR of order four with Minnesota prior over 113 quarterly observations from 1998 Q1 to 2026 Q1, with the US federal funds rate and China's GDP growth entering as block-exogenous forcing variables. US rate conditions pass through strongly to HIBOR under the currency board, consistent with LERS arbitrage. GDP propagation is amplified mainly through the property market: property price shocks explain 20.7% of GDP forecast error variance at an eight-quarter horizon, exceeding the direct trade channel (exports: 16.4%) and direct monetary channel (HIBOR: 8.4%). The direct HIBOR–GDP link is more persistent than the property sub-channel, remaining significant through four quarters. Recursive local projections support the fast HIBOR–property and property–GDP timing; the exports–GDP channel is significant in the BVAR but weakens in LP after conditioning on current China GDP growth. Out-of-sample RMSE comparisons serve as model discipline only. Recursive identification and residual autocorrelation limitations are stated explicitly in Section 7.

1 Introduction

A currency board eliminates the exchange rate as a shock absorber and hands domestic interest rate setting to the anchor country. Under Hong Kong's Linked Exchange Rate System, the three-month interbank offered rate (HIBOR) has tracked the US federal funds rate since the peg's establishment in 1983, absorbing US monetary cycles that Hong Kong's own authorities cannot offset. When the Federal Reserve tightens, Hong Kong's interbank rate rises in step; yet Hong Kong cannot lower rates to cushion the domestic impact. What remains is the question this paper addresses: through which domestic channels do imported interest rate movements transmit to output, and with what speed and persistence?

We answer this question using a Bayesian VAR of order four (BVAR(4)) estimated on 113 quarterly observations from 1998 Q1 to 2026 Q1. The model includes six endogenous variables—the three-month HIBOR, exports to China, property prices, GDP growth, CPI inflation, and the unemployment rate—with the US federal funds rate and China's GDP growth entering as strictly exogenous forcing variables, following the block-exogenous structure of Cushman and Zha (1997). Structural shocks are identified via a Cholesky decomposition whose ordering reflects institutional timing under the currency board. The Minnesota prior regularises inference over a sample that spans the Asian financial crisis, the global financial crisis, and the COVID-19 pandemic.

The main results are as follows. First, the estimated contemporaneous coefficient on `us_ffr` in the HIBOR equation is $+0.511$ ($p < 0.001$), consistent with strong LERS pass-through; HIBOR's 95% own-share in its FEVD reflects that domestic shocks account for little of its variation. Second, a HIBOR tightening depresses property prices significantly within one quarter (90% credibility interval: $[-0.967, -0.316]$) but the response crosses zero by $h = 2$. Third, despite this short-lived property response, property price shocks are the dominant source of GDP variance at the eight-quarter horizon (20.7%), larger than the direct trade channel (exports: 16.4%) or direct monetary channel (HIBOR: 8.4%). Fourth, the direct HIBOR–GDP link is significant at horizons one through four (90% CI at $h = 4$: $[-0.296, -0.067]$), indicating a credit and mortgage sub-channel that sustains output contraction beyond the asset price effect. Fifth, exports to China explain 16.4% of GDP forecast error variance in the BVAR and are significant at $h = 1$ and $h = 2$; in recursive LP, this channel weakens after conditioning on current China GDP growth.

Two limitations bear on interpretation. `us_ffr` is an external monetary-conditions variable, not a structural policy surprise; results should be read as reflecting US rate cycles, not identified Fed shocks. The recursive local projections impose the same Cholesky ordering as the BVAR and serve as a robustness check on direct-projection timing, not as an alternative identification strategy.

The paper makes two contributions. First, it documents a speed asymmetry within monetary transmission under a currency board: the property sub-channel is fast (significant at $h = 1$–2) but short-lived, while the direct credit sub-channel is slower but more persistent (significant through $h = 4$). Mishkin (1996) notes that interest rate and asset price channels operate on different timescales; our FEVD and IRF evidence quantifies this asymmetry under the institutional conditions of LERS. Second, the paper shows that domestic property amplification accounts for a larger share of GDP forecast error variance (20.7%) than either the direct monetary channel (8.4%) or the trade channel (16.4%) at an eight-quarter horizon, under recursive identification with HIBOR ordered first.

The remainder of the paper is organised as follows. Section 2 describes the institutional context. Section 3 describes the data. Section 4 presents the empirical strategy. Section 5 presents the main results. Section 6 reports robustness checks. Section 7 states limitations. Section 8 concludes.

2 Institutional Background and Channels

Hong Kong has operated a currency board since October 1983, with the Hong Kong dollar fixed at HKD 7.80 per US dollar. The LERS self-corrects through interest rate arbitrage: capital inflows expand the monetary base and compress HIBOR toward the federal funds rate; outflows drain the base and push HIBOR upward. The mechanism is automatic and operates within days, so over quarterly horizons HIBOR is mechanically tied to the federal funds rate. Hong Kong cannot engineer a real depreciation to offset a US tightening cycle; what remain are the domestic transmission channels.

Three domestic channels are relevant. The *credit and property channel*: HIBOR increases raise mortgage costs and compress property valuations, tightening collateral constraints and investment (Bernanke and Blinder, 1992; Goodhart and Hofmann, 2008). In Hong Kong, residential property is the primary household asset and dominant loan collateral, concentrating this channel in a single highly liquid asset class. The *direct credit channel*: HIBOR also raises business financing costs directly, sustained beyond any property price adjustment through contract renegotiation and new loan origination. The *trade channel*: Hong Kong's economy is deeply integrated with mainland China, which by the mid-2010s accounted for over half of total re-exports. China's growth cycle drives export demand independently of the monetary channel.

The block-exogenous structure follows Cushman and Zha (1997): the small open economy cannot contemporaneously influence the foreign interest rate or China's output, and Hong Kong's monetary base is too small to perturb US monetary conditions. Under LERS, this is an institutional constraint, not merely a plausibility argument.

3 Data

The dataset covers 113 quarterly observations from 1998 Q1 to 2026 Q1. The start date is regime-justified: the 1997 handover and Asian financial crisis constitute a structural break in LERS credibility and Hong Kong–China trade integration. The three-month HIBOR panel is continuous across the handover (the same HKAB survey panel has operated since the mid-1980s), so no instrument break forces the 1998 start; the decision is economic. Table 1 summarises the eight variables.

Property prices enter as quarter-on-quarter percentage changes: the level index is I(1) (ADF $p = 0.82$) while the QoQ change is stationary (ADF $p < 0.001$). HIBOR and unemployment enter in levels, as both are mean-reverting under the peg and through

Table 1: Variable Dictionary

Variable	Role	Form	Units	Source
`us_ffr`	Exogenous	Level	% p.a.	FRED FEDFUNDS, monthly → quarterly mean
`china_gdp`	Exogenous	YoY %	%	OECD QNA `CHN.B1_GE.GPSA.Q`
`hibor_3m`	Endogenous	Level	% p.a.	HKMA hibor.fixing, monthly → quarterly mean
`hk_exports_china_yoy`	Endogenous	YoY %	%	C&SD table 410-50013, nominal HKD
`hk_property_price_qoq`	Endogenous	QoQ %	%	RVD All Classes index, quarterly differenced
`gdp_growth`	Endogenous	YoY %	%	C&SD table 310-30001, real chain-linked
`cpi_inflation`	Endogenous	YoY %	%	C&SD table 510-60001, Composite CPI
`unemployment`	Endogenous	Level	%	C&SD table 210-06101, seasonally adjusted M3M

normal labour market adjustment respectively. Unemployment, CPI, and the property price level are confirmed I(1); ADF and KPSS tests on the remaining endogenous variables support stationarity or near-stationarity. Johansen trace and max-eigenvalue tests on the endogenous I(1) block find rank zero at the 95% level, so no cointegrating relationship exists and a VECM is not warranted (Sims et al., 1990).

Three data limitations are noted. First, exports are measured in nominal Hong Kong dollars; volume and price effects cannot be separated. Second, `us_ffr` is a monetary-conditions proxy rather than a clean policy surprise. Third, the post-2010 China–Hong Kong financial integration channel is not directly modelled as no suitable quarterly series extends back to 1998.

4 Empirical Strategy

4.1 BVARX Specification

We estimate a Bayesian VAR of order four,

$$\mathbf{y}_t = \mathbf{c} + \sum_{\ell=1}^{4} \mathbf{A}_\ell \, \mathbf{y}_{t-\ell} + \mathbf{B}_0 \, \mathbf{z}_t + \boldsymbol{\varepsilon}_t, \qquad \boldsymbol{\varepsilon}_t \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma}), \tag{1}$$

where $\mathbf{y}_t$ is the 6×1 vector of endogenous variables and $\mathbf{z}_t = (\texttt{us\_ffr}_t, \texttt{china\_gdp}_t)'$ is the block-exogenous forcing vector. The model has $k = 27$ parameters per equation and

$T_{\text{eff}} = 109$ observations after conditioning on four initial lags.

Prior. The Minnesota prior of Litterman (1986) shrinks lagged own values toward a random walk with harmonic decay $1/\ell^2$ across lags, and shrinks cross-variable lags toward zero. The tightness parameter $\pi_1 = 0.085$ and cross-variable weight $\pi_2 = 1.0$ are marginal-likelihood optimal, estimated jointly with the AR prior means δ_i via Giannone et al. (2015). The optimal prior means range from $\delta = 0.418$ (property prices) to $\delta = 0.991$ (unemployment), calibrated to each series' integration order. The prior on exogenous coefficients $\mathbf{B}_0$ is diffuse ($\pi_4 \approx 100$).

Posterior and inference. With $\mathbf{\Sigma}$ fixed at AR residual variances, the Minnesota prior yields a closed-form Normal posterior; all draws are independent and identically distributed. Credibility bands are 90% posterior intervals from 3,000 independent draws. No Markov chain Monte Carlo simulation is employed: the analytical posterior makes trace-plot and effective-sample-size diagnostics meaningless here. At $h = 1$, the Cholesky impact matrix is deterministic given $\mathbf{\Sigma}$, so posterior bands have zero width; $h = 2$ is the first horizon with genuine posterior uncertainty.

Exogenous lag structure. Exogenous variables enter contemporaneously only (VARX(4,0)) following Cushman and Zha (1997). For `us_ffr`, the currency board arbitrages HIBOR to the federal funds rate within the quarter, so lagged FFR values are spanned by current HIBOR and its own lags. For `china_gdp`, the year-on-year construction shares three of four underlying quarters between adjacent observations, generating near-collinearity with the lagged value.[1] Adding $\texttt{us\_ffr}_{t-1}$ does not remove Ljung-Box failures in the GDP and CPI equations; residual autocorrelation reflects structural instability rather than an omitted lag. This is consistent with Cushman and Zha (1997)'s block-exogeneity framework and confirms that $q = 0$ is the appropriate exogenous lag order.

4.2 Recursive Identification

Structural shocks are identified via a recursive (Cholesky) decomposition with ordering: HIBOR, exports, property, GDP, CPI, unemployment. HIBOR is ordered first: under LERS, the interbank rate is mechanically determined by the federal funds rate and cannot respond to domestic output or property conditions within the same quarter. This is an institutional constraint, not a maintained statistical assumption. Exports are ordered second: driven by mainland China demand, they are contemporaneously exogenous to domestic monetary and price conditions. Property is ordered third, ahead of GDP: under

[1] The 75% semantic overlap between adjacent year-on-year China GDP observations means including $\texttt{china\_gdp}_{t-1}$ adds a near-duplicate regressor. Identification of a genuine trade-lag effect would require quarter-on-quarter China GDP data, which are unavailable for the full sample.

LERS, property prices are the primary within-quarter transmission vehicle for monetary conditions, and asset price adjustment precedes measured GDP revision in quarterly national accounts.[2] CPI and unemployment are ordered last, reflecting standard price and labour market sluggishness.

FEVD estimates use recursive identification by construction; the local projections check whether the same domestic channel timing appears under direct projections with HAC inference and the same BVARX information set.

4.3 Local Projections

We estimate local projections (Jordà, 2005) for each of the four domestic channels as a timing robustness check (Ramey and Zubairy, 2018). The LP regression at horizon h is

$$y_{t+h} = \alpha_h + \beta_h \, s_t + \boldsymbol{\gamma}_h' \mathbf{x}_t + \varepsilon_{t+h}, \tag{2}$$

where s_t is the shock variable and $\mathbf{x}_t$ includes (i) four lags of all endogenous variables, (ii) current-period values of the exogenous block ($\texttt{us\_ffr}_t$, $\texttt{china\_gdp}_t$), and (iii) contemporaneous values of all endogenous variables ordered before s_t in the Cholesky decomposition. Including controls in (iii) imposes the same recursive identifying restrictions as the BVAR, so $\hat{\beta}_h$ has the same structural interpretation at each horizon. The LP does not remove recursive identification; it provides a robustness check on domestic channel timing under direct estimation with HAC-robust inference. Newey-West standard errors with bandwidth $h + 1$ are used throughout.

5 Results

5.1 External Forcing Check

The HIBOR own-share in its own forecast error variance is 95% at an eight-quarter horizon. This reflects near-mechanical pass-through of US monetary conditions under the currency board: within any quarter, HIBOR is driven by `us_ffr` and peg credibility, not by domestic output or property conditions. Exogenous dynamic multipliers—recovered from the block-exogenous impulse response object—show that a one percentage point increase in the federal funds rate raises Hong Kong GDP by +0.225 pp at $h = 1$ (US demand correlation) before turning negative at $h = 4$ (-0.199 pp) as monetary tightening propagates. A one percentage point increase in China GDP growth raises HK GDP by +0.216 pp at $h = 1$ and +0.128 pp at $h = 2$. The positive FFR–GDP sign at $h = 1$ is a standard open-economy result: contemporaneous US demand conditions dominate within-

[2] This ordering is load-bearing for the 20.7% FEVD share. A reverse ordering — GDP before property — is the standard diagnostic robustness check and is left for future work.

quarter output before the tightening effect propagates (Cushman and Zha, 1997). These end-to-end multipliers complement the FEVD; they are not structural decompositions.

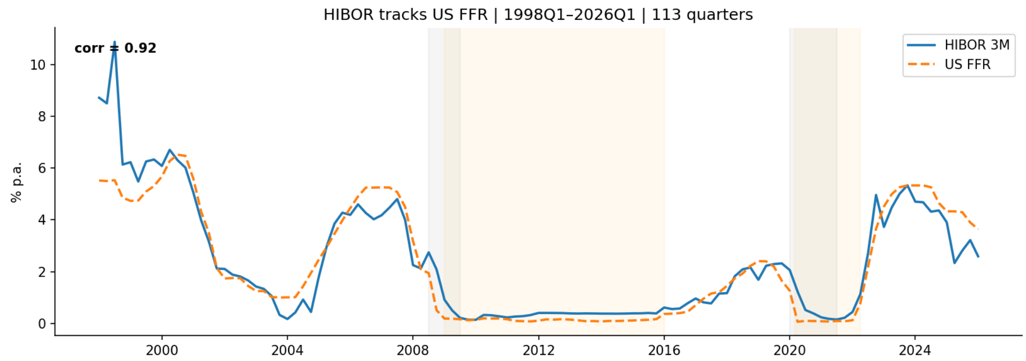

Figure 1: HIBOR and US Federal Funds Rate, 1998Q1–2026Q1

5.2 Main Impulse Responses

Table 2 reports the four key transmission channels at horizons $h \in \{1, 2, 4\}$ quarters. Figure 2 presents the full 6×6 impulse response matrix.

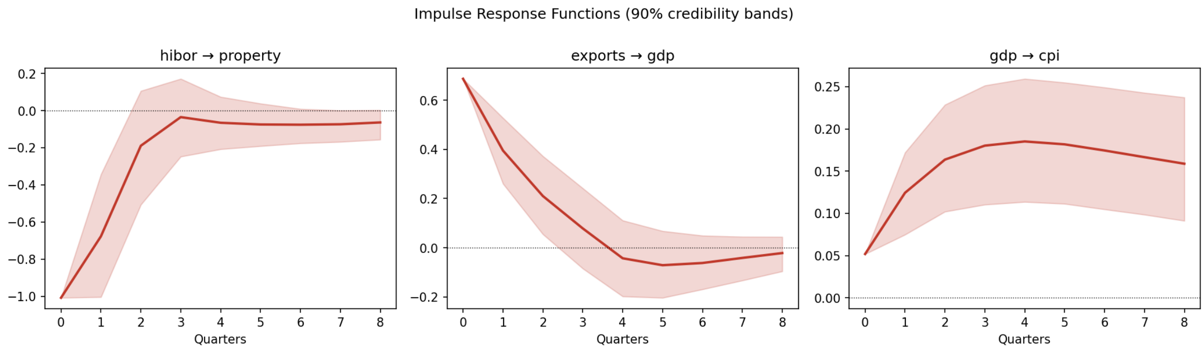

Figure 2: Canonical BVAR(4) Impulse Response Functions, 90% Credibility Bands

Channel 1: monetary channel. A HIBOR tightening depresses property prices significantly within one quarter (90% CI: $[-0.967, -0.316]$), consistent with the collateral channel described by Goodhart and Hofmann (2008). The property price response crosses zero at $h = 2$ and is not significant beyond the first quarter. This short-lived property response coexists with a direct HIBOR–GDP link significant at $h = 1$, $h = 2$, and $h = 4$. Together with the property–GDP IRF (significant at $h = 1$ and $h = 2$), this establishes the speed asymmetry: a *fast property sub-channel* (significant at $h = 1$–2) and a *slow direct sub-channel* (persistent through $h = 4$ via credit costs and mortgage rates). The BVAR significance at $h = 4$ for the direct channel should be treated with caution given that the recursive LP does not confirm it at that horizon.

Table 2: Canonical Impulse Responses: 90% Posterior Credibility Bands

Channel	h	Lower	Upper	Significant?
HIBOR → Property	1	−0.967	−0.316	Yes
	2	−0.537	+0.066	No
	4	−0.229	+0.076	No
HIBOR → GDP (direct)	1	−0.408	−0.057	Yes
	2	−0.493	−0.130	Yes
	4	−0.296	−0.067	Yes
Property → GDP	1	+0.330	+0.634	Yes
	2	+0.247	+0.580	Yes
	4	—	—	No
Exports → GDP	1	+0.266	+0.540	Yes
	2	+0.060	+0.375	Yes
	4	−0.188	+0.121	No
GDP → CPI	1	+0.076	+0.174	Yes
	2	+0.104	+0.231	Yes
	4	+0.117	+0.264	Yes

Notes: BVAR(4), Minnesota prior. Responses to a one standard deviation structural shock identified by Cholesky decomposition. At $h = 1$ the Cholesky impact matrix is deterministic given fixed $\boldsymbol{\Sigma}$; the posterior band degenerates to a point and significance refers to the sign of that point response. At $h = 4$ for Property → GDP, the band crosses zero; point estimate and bound values are omitted. Significance means the 90% credibility interval excludes zero.

Channel 2: trade channel. A positive China GDP growth shock raises exports and transmits to Hong Kong GDP at $h = 1$ (90% CI: $[+0.266, +0.540]$) and $h = 2$ (90% CI: $[+0.060, +0.375]$) before becoming insignificant at $h = 4$. Exports–GDP is a BVAR channel: significant at $h = 1$–2 and accounting for 16.4% of GDP FEVD at $h = 8$. In recursive LP, it becomes weak after conditioning on current China GDP growth. Because $\texttt{china\_gdp}_t$ and exports are near-collinear, the LP's single-equation projection absorbs most identifying variation into the exogenous control, leaving the exports coefficient imprecise; the BVAR system estimation partially preserves this through joint prior regularisation across equations.

Inflation dynamics. GDP–CPI transmission is significant and persistent at $h = 1$, $h = 2$, and $h = 4$, consistent with standard demand-pull inflation dynamics. The CPI equation exhibits a COVID-19 mean break ($p = 0.034$; Section 6.2), so inflation transmission estimates from 2020–2021 carry additional uncertainty.

5.3 GDP Forecast Error Variance Decomposition

Table 3 reports the FEVD of GDP growth at an eight-quarter horizon. Figure 3 displays the full decomposition for all six equations. Figure 4 focuses on the GDP equation.

Table 3: Forecast Error Variance Decomposition: GDP Growth at $h = 8$

Shock source	Share of GDP variance (%)
Property prices	20.7
Exports to China	16.4
HIBOR	8.4
GDP own + other	54.5
Total	100.0

Notes: BVAR(4), Minnesota prior, HIBOR-first Cholesky ordering. FEVD covers endogenous recursive shocks only; external-variable contributions (FFR, China GDP) are absorbed into HIBOR and exports own-shocks respectively. CPI and unemployment shock contributions are individually small and are grouped into the residual row. HIBOR's own-share in its own forecast error variance is 95%, consistent with near-mechanical peg dynamics.

Property shocks dominate GDP variance (20.7%) despite the HIBOR–property impulse response being significant for only one quarter. This reflects the collateral amplification mechanism: a property price decline compresses collateral values and tightens credit conditions across several subsequent quarters, generating cumulative output variance beyond the initial price impact (Goodhart and Hofmann, 2008). The direct trade channel (16.4%) reflects the dominance of mainland China as Hong Kong's primary trading partner. The direct monetary channel (8.4%) is smaller in variance terms but more persistent in impulse response dynamics.

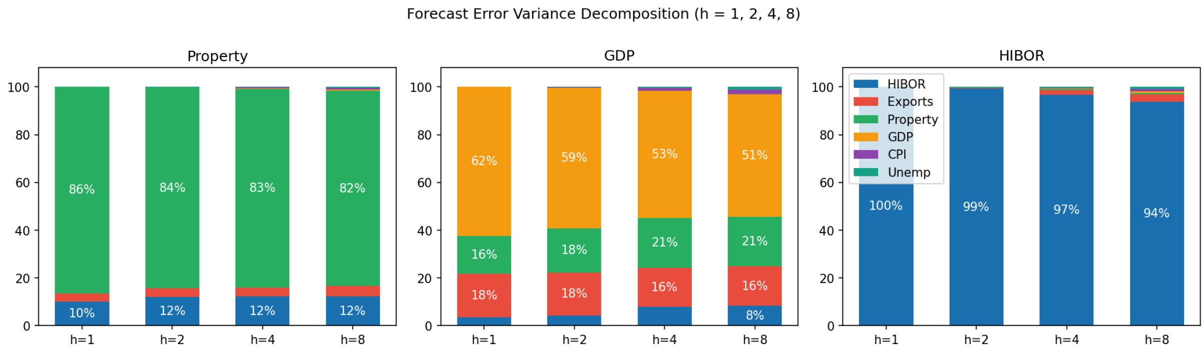

Figure 3: Forecast Error Variance Decomposition, All Equations

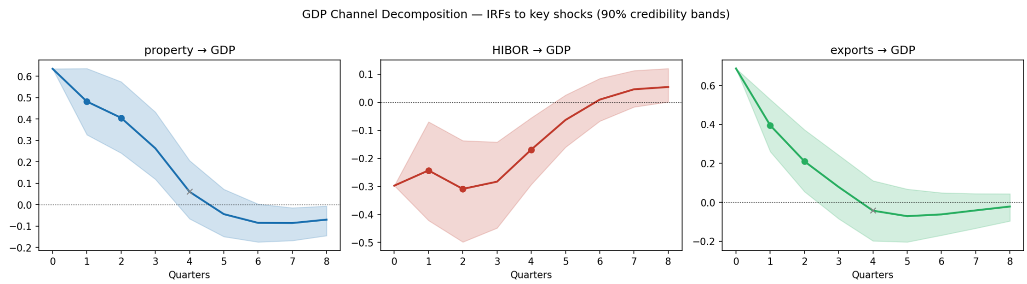

Figure 4: GDP Channel Decomposition: Accumulated FEVD by Shock Source

6 Robustness and Diagnostics

6.1 Recursive LP Timing Check

Table 4 compares LP and BVAR significance for the four domestic channels. Three of four channels are confirmed. The HIBOR–property and property–GDP channels are consistent across both methods at the key horizons. HIBOR–GDP is confirmed at $h = 1$ and $h = 2$; the LP result at $h = 4$ is not significant, and the BVAR's $h = 4$ finding should be treated as suggestive. Exports–GDP is not significant in LP at any horizon after conditioning on current China GDP growth. Figure 5 presents the LP estimates alongside BVAR credibility bands.

6.2 Structural Stability

Table 5 reports mean break (Welch t-test) and variance shift (Levene test) results on BVAR(4) residuals at four break dates: SARS (2003 Q2), GFC (2008 Q3), China slowdown (2015 Q3), and COVID-19 (2020 Q1). These are proper hypothesis tests with null distributions; penalised segmentation algorithms are not used.

SARS 2003 and China 2015 leave no statistical trace in any equation. The GFC generates variance shifts in HIBOR ($p = 0.037$) and exports ($p = 0.003$) residuals but no GDP mean break ($p = 0.140$). COVID-19 produces a mean break in CPI only ($p = 0.034$) and variance shifts in exports ($p = 0.029$) and unemployment ($p = 0.002$). GDP shows

Table 4: LP Robustness: Channel Significance at Key Horizons

Channel	h	LP significant (90% HAC)?	BVAR significant (90% CI)?
HIBOR → Property	1	Yes	Yes
	2	No	No
HIBOR → GDP	1	Yes	Yes
	2	Yes	Yes
	4	No	Yes
Exports → GDP	1	No	Yes
	2	No	Yes
Property → GDP	1	Yes	Yes
	2	Yes	Yes

Notes: LP uses the same BVARX information set with contemporaneous recursive controls. HAC bandwidth $h+1$ at each horizon. The exports–GDP channel is not significant in LP after conditioning on current China GDP growth; the BVAR recovers this channel through cross-equation shrinkage.

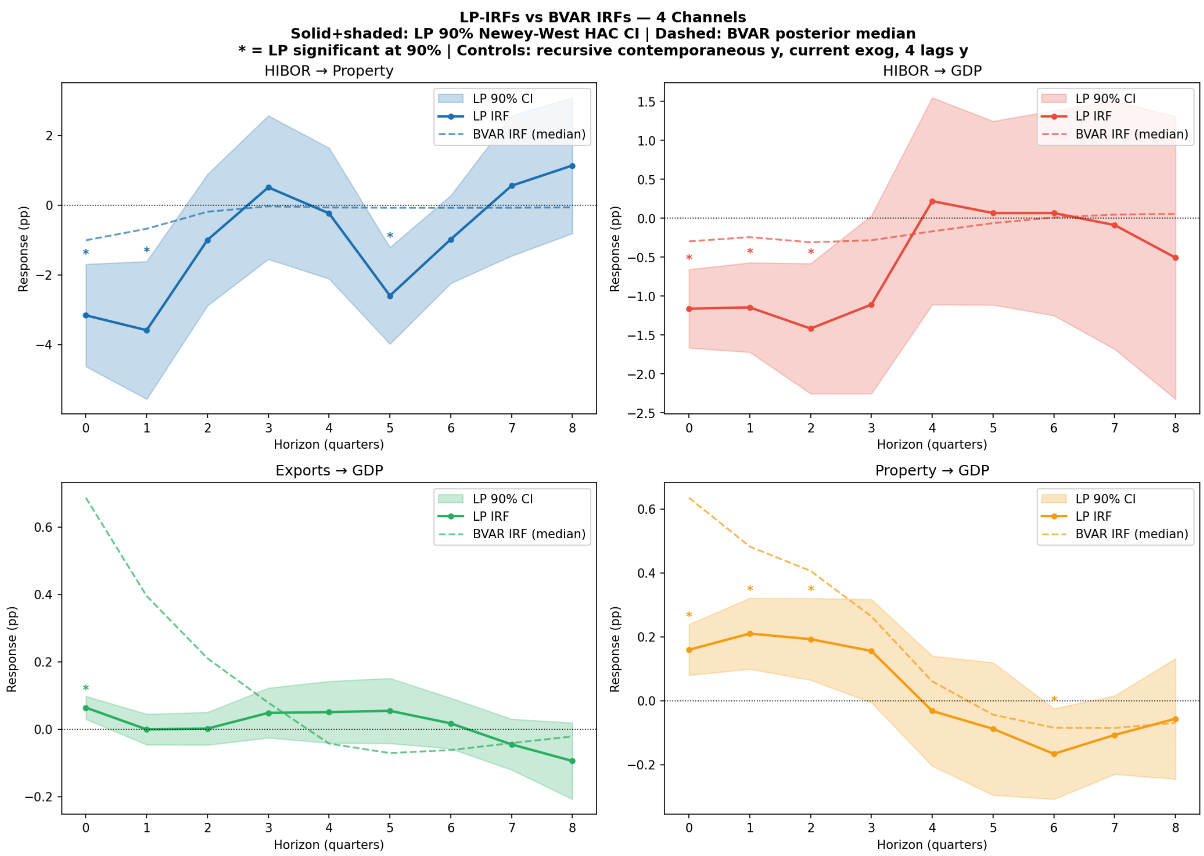

Figure 5: Local Projection vs. BVAR: Impulse Responses for Four Channels

no mean break at either the GFC or COVID-19. The GDP break that appeared under VARX(1) at the GFC ($p = 0.006$) disappears under BVAR(4), consistent with Minnesota shrinkage absorbing apparent parameter instability from insufficient lag structure.

Table 5: Structural Stability Tests on BVAR(4) Residuals

Equation	SARS 2003	GFC 2008	China 2015	COVID 2020
hibor	stable	var. shift ($p = 0.037$)	stable	stable
exports	stable	var. shift ($p = 0.003$)	stable	var. shift ($p = 0.029$)
property	stable	stable	stable	stable
gdp	stable	stable ($p = 0.140$)	stable	stable ($p = 0.264$)
cpi	stable	stable	stable	**mean break** ($p = 0.034$)
unemployment	stable	stable	stable	var. shift ($p = 0.002$)

Notes: "stable" indicates failure to reject the null of no break at the 10% level. Bold entry marks the one limitation of the fixed-coefficient specification. GDP p-values shown for the two crisis dates to emphasise the absence of a break despite large shocks.

Variance shifts at COVID-19 reflect larger shocks through an otherwise stable transmission architecture, not a change in the mechanism itself.

6.3 Zero Lower Bound Asymmetry

We test whether HIBOR–property transmission is asymmetric across interest rate regimes. Let $\text{ZIRP}_t = 1$ if $\texttt{us\_ffr}_t < 0.25\%$, capturing the US near-zero rate environment from 2009 Q1 to 2022 Q1 (36 of 113 observations). A threshold LP-IRF at $h = 1$ yields:

$$\begin{aligned} \hat{\beta}_{\text{normal}} &= -2.675 \quad 90\% \text{ CI: } (-3.826,\ -1.523) \quad [\text{significant}], \\ \hat{\beta}_{\text{ZIRP}} &= -0.604 \quad 90\% \text{ CI: } (-7.704,\ +6.496) \quad [\text{not significant}]. \end{aligned}$$

The directional pattern is consistent with impaired property transmission at the zero lower bound. The ZIRP estimate is imprecisely estimated given the limited subsample; the asymmetry is directionally confirmed but not conclusively established.

6.4 Unemployment Specification

Unemployment is retained in levels despite marginal evidence of a unit root (ADF $p = 0.091$). Re-estimating with first-differenced unemployment leaves the HIBOR–property and exports–GDP channels qualitatively unchanged at $h = 2$, with band shifts below 0.1 percentage point and no change in significance. We retain the level specification following Sims et al. (1990).

6.5 Exogenous Lag Sensitivity

Adding one lag of `us_ffr` (VARX(4,1), $k = 28$ parameters per equation) leaves Ljung-Box statistics essentially unchanged across GDP and CPI equations. Impulse responses for the HIBOR–property, exports–GDP, and property–GDP channels are stable across specifications. We retain VARX(4,0) as the baseline.

6.6 Out-of-Sample Forecast Performance

This is not mainly a forecasting paper. The BVAR(4) is evaluated against VARX(1) as model discipline using recursive out-of-sample RMSE ratios (BVAR/VARX1, < 1 indicates BVAR superiority). The BVAR wins in 18 of 18 cells across six variables and three horizons ($h \in \{1, 2, 4\}$). Selected ratios: exports at $h = 1$: 0.813; GDP at $h = 1$: 0.942; CPI at $h = 1$: 0.907. The few $h = 4$ ratios above 1.0 are all within 0.02 of parity. Forecast evaluations are conditional: realised paths of `us_ffr` and `china_gdp` are provided over the forecast window, consistent with the strict exogeneity assumption.

7 Limitations

Five limitations bound the scope of inference from this paper.

Recursive identification is order-dependent. All IRF and FEVD results are conditional on the Cholesky ordering HIBOR, exports, property, GDP, CPI, unemployment. The LERS mechanism provides unusually strong institutional support for placing HIBOR first. The remaining ordering decisions are economically motivated but not uniquely determined. Readers who prefer a different contemporaneous structure would obtain different channel shares.

`us_ffr` is not a structural monetary policy surprise. The model uses the quarterly average federal funds rate as an external monetary-conditions variable. This series conflates anticipated and unanticipated policy changes, term premium variation, and global risk factors. No Romer-Romer or high-frequency identification is applied. Results should be interpreted as responses to imported financial conditions under LERS, not to identified US monetary policy shocks.

FEVD does not allocate external-variable variance. The FEVD reported in Table 3 covers only the endogenous recursive shock decomposition. The contributions of `us_ffr` and `china_gdp` to GDP variance are not directly visible in the FEVD; they are partially absorbed into the HIBOR and exports own-shocks respectively. The exogenous dynamic multipliers in Section 5 provide end-to-end external scaling, but these are not orthogonal structural decompositions.

GDP and CPI residual autocorrelation remains. Ljung-Box portmanteau tests reject the null of white-noise residuals in the GDP and CPI equations. Adding one lag of `us_ffr` does not resolve this. The autocorrelation co-locates with the GFC and COVID-19 windows, suggesting episodic structural instability rather than persistent misspecification. LP inference in Section 6.1 is valid under this serial correlation by construction (HAC standard errors at bandwidth $h + 1$). BVAR point estimates for the affected equations carry additional uncertainty.

Sample spans major structural breaks. The 28-year sample covers the Asian financial crisis aftermath, the GFC, and COVID-19. The CPI equation has a confirmed COVID-19 mean break ($p = 0.034$). The fixed-coefficient Minnesota prior partially absorbs this through shrinkage but cannot fully accommodate discrete regime changes. Inflation transmission results from 2020–2021 are unreliable. Future work with a longer post-COVID sample could apply time-varying parameter methods to formally assess whether the transmission architecture evolved across the 2022–2023 normalisation cycle.

8 Conclusion

This paper estimated a BVAR(4) with Minnesota prior on 113 quarterly observations of Hong Kong macroeconomic data, using the US federal funds rate and China's GDP growth as block-exogenous forcing variables. The institutional features of the LERS provide unusually sharp identification for the first step of the Cholesky ordering: HIBOR is mechanically determined by the federal funds rate within the quarter, validating its position as the most exogenous domestic variable.

The main finding is a speed asymmetry within the monetary transmission channel. HIBOR shocks transmit rapidly to property prices within one quarter but the property price response is not persistent beyond that horizon. The direct HIBOR–GDP link persists through four quarters via credit and mortgage adjustment. Property price shocks are the dominant source of GDP variance at an eight-quarter horizon (20.7%), exceeding the direct trade channel (16.4%) and direct monetary channel (8.4%). Recursive LP with HAC-robust standard errors confirms domestic channel timing for three of four channels; the exports–GDP channel is significant in the BVAR but weakens in LP after conditioning on current China GDP growth. A threshold LP analysis indicates monetary transmission is attenuated in the ZIRP regime, though this result is imprecisely estimated.

The results highlight that domestic property amplification, not imported rate movements directly, is the principal source of GDP forecast error variance in this currency board economy. Because property is the largest single source of GDP forecast-error variance (20.7%), macroprudential tools act on the dominant domestic variance channel. The trade channel result establishes that China's growth cycle is a substantial secondary driver of Hong Kong output variance.

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